WÖHLER

Calibration Instructions
Flue Gas Analyzer

Wöhler A 550
Wöhler A 550 L

Art, no. 23431– 2014-02-20

The Measure of Technology

Contents

1 Information about this manual

This manual is aimed at authorized control and service points only. Do not pass this information to others.

2 Technical Data

2.1 Readings

Oxygen (O_2) concentration in flue gas	
Reading	% of flue gas volume (dry conditions)
Measurement principle	electrochemical sensor
Range	0.0 to 21.0 %
Accuracy	± 0.3 % according to VDI 4206 page 1
Carbon monoxide (CO_V 4000 ppm) in flue gas (only Wöhler A 550)	
Reading	ppm of flue gas volume (dry conditions)
Measurement principle	Electrochemical sensor, H_2 compensated
Range	0 to 4,000 Vol. ppm resolution 1 Vol.-ppm
Accuracy	± 20 ppm (< 400 ppm), otherwise 5% of reading according to VDI 4206 p1
Carbon monoxide (CO_V) in flue gas	
Reading	ppm of flue gas volume (dry conditions)
Measurement principle	electrochemical sensor not H_2 compensated
Range	0 to 35,000 Vol. ppm resolution 1 Vol.-ppm
Accuracy	± 100 ppm (< 1,000 ppm), otherwise 10% of reading (when H_2 < 5 % of reading)

Hydrogen (H_2 2,000 ppm) concentration in flue gas	
Reading	ppm of flue gas volume (dry conditions)
Measure-ment principle	electrochemical sensor
Range	0 to 2,000 Vol. ppm resolution 1 Vol.-ppm (between 15° and 40°)
Accuracy	± 40 ppm (< 400 ppm), other-wise 10% of reading
Carbon monoxide (CO_V high) in flue gas (option)	
Reading	ppm of flue gas volume (dry conditions)
Measure-ment principle	electrochemical sensor
Range	0 to 100,000 Vol. ppm resolution 1 Vol.-ppm
Accuracy	± 100 ppm (< 1,000 ppm), otherwise 10% of reading (when H_2 < 5 % of reading)

Nitrogen monoxide (NO_V) in flue gas (option)	
Reading	ppm of flue gas volume (dry conditions)
Measurement principle	electrochemical sensor
Range	0 to 3,000 Vol.-ppm, permanently up to 1,000 ppm resolution 1 Vol.-ppm
Accuracy	± 5 Vol.-ppm (< 100 ppm), otherwise 5 % of reading
Nitrogen monoxide (NO_2) in flue gas (option)	
Reading	ppm of flue gas volume (dry conditions)
Measurement principle	electrochemical sensor
Range	0 to 1,000 Vol.-ppm, (permanently up to 200 ppm) resolution 1 Vol.-ppm
Accuracy	± 5 Vol.-ppm (< 100 ppm), otherwise 5 % of reading
Draft (PD) with 4 Pa Test (only Wöhler A 550)	
Reading	Pascal
Measurement principle	Semiconductor
Range	0.00 to ± 110.00 hPa, resolution 0.1 Pa (<1000.0 Pa), otherwise 1 Pa, 0.01 Pa for the ventilation measurement
Accuracy	0.3 Pa (<10Pa), otherwise 3% of reading Drift < 0.2 Pa in 5 minutes

Draft (PD)	
Reading	Pascal
Measurement principle	Semiconductor
Range	0.00 to ± 110.00 hPa, resolution 1 Pa
Accuracy	2 Pa (< 40Pa), otherwise 5% of reading
Flue gas temperature (TA)	
Reading	°C
Measurement principle	Thermocouple (NiCr-Ni)
Range	-20.0 °C to 800.0 °C resolution 0.1 °C
Accuracy	0 - 133 °C: ± 2°C 133 - 800 °C: 1.5% of reading
Ambient Air Temperature (TL)	
Reading	°C
Measurement principle	Thermocouple (NiCr-Ni)
Range	-20.0 °C to 100 °C resolution 0.1 °C
Accuracy	± 1°C

Wood moisture	
Reading	water mass related to the absolute dry fuel mass
Measurement principle	resistance measurement
Range	10.0 to 40.0 %, resolution 0.1 %
Accuracy	40 % of reading according to VDI 4206 page 4
Wood temper-ature	5 - 25 °C
Lifetime of the electrodes	Depending on the frequency of utilization The electrodes will work cor-rectly, if there is no surface damage or bending.

2.2 Calculated Values

Calculated Value	Explanation
Q_A	Losses in % depending on the fuel
ETA	Efficiency (0.0 to 120%)
CO_2 in Vol. -%	Range 0 – CO_{2max}, resolution 0.1 %
CO_n	Average CO content ($CO_{standard}$) in relation to the adjustable reference oxygen value Default: 0% (oil and gas).
U-value	Heat transmission coefficient W/(m^2K)
Dew Point in the flue gas	°C
Medium soot level	Soot number ± 0.1
Excess air coefficient	Lambda λ (e.g. 1.25 when the excess of air is 25%
Condensate quantity	kg/m^3 gas or kg/kg fuel
Air velocity	0.1 to 130 m/s, resolution < 0.1 m/s, for the measurement of the ventilation loss (heating check)

2.3 Technical Data

Description	Data
Power supply	Lithium-ion battery 3.7 V, 5800 mAh, charging via USB
Working time	Ca. 12 h (depending on the operating mode and the Display Back-light)
Charging time for a full charge	Approx. 7 hours
Storage Temperature	-20 °C to 50 °C
Operation temperature	+5 °C to 40 °C to guaranty the specified accuracy
Relative humidity	30% to 70 %
Weight	1250 g
Dimensions	220 x 160 x 55 mm (without probe)
Length of the hose	1700 mm

3 Six-monthly check

The analyzer has to be checked every 6 month at a service point which is officially authorized. The prescription VDI 4208, page 2 specifies the minimum requirements.

The following tests have to be performed:

- with test gas: O_2 measurement, CO measurement, NO measurement (option)
- The flue gas temperature T_A has to be checked with a reference standard at two measuring points: one point in the lower part and one point in the upper part of the measuring range.
- The ambient air temperature T_L has to be checked with a reference standard at one measuring point.
- The draft has to be checked with an adequate draft regulator.
- The flow of the flue gas to be analyzed must be controlled by a rotameter at the sucking side of the gas probe.
- The gas probe and the analyzer have to be optically checked for pollution, condensation etc.
- Control of the measuring channel for wood moisture with a reference resistance. According to VDI 4206, page 4 the displayed wood moisture may not differ more than ± 5% from the reference value.

4 Calibration and adjustment

4.1 Enter calibration mode

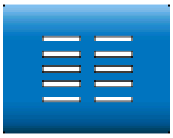
- If necessary carry out a firmware update of the analyzer. Use the PC software Wöhler A 550 to do so. After the update the analyzer will start again.

Diagnosis menu

- Click on the "diagnosis" icon to finish the core flow search and enter the analyzer's self diagnosis mode which gives you information about the unit and the sensors.

- Close the diagnosis menu and wait until the search for the core flow has finished.

Main menu



- Go to the main menu and click on the calibration icon.

ATTENTION!
Only trained persons are allowed to calibrate the analyzer. Improper changes may lead to incorrect results.

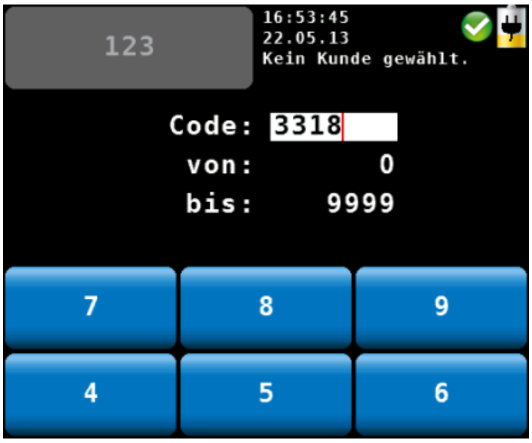


Fig. 1: Entering the access code

The calibration menu is protected by an access code, so that only authorized personal can calibrate the analyzer.

- Enter the access code **3318** and confirm with "OK".

4.2 Factory calibration

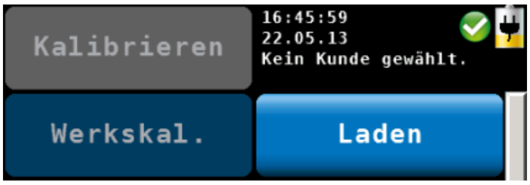


Fig. 2: Restore the factory calibration

- Click on "Werkskalibrierung laden" to restore the factory calibration. We recommend to do so if the last calibration has not been carried out correctly.

4.3 ZIV number

Fig. 3: Entering the ZIV number

- To enter the identification number of the unit, click on the field at the right of the ZIV number. The type designation and the serial number are fix. Enter the code of the service point. The date will update automatically.

4.4 Calibration of the temperature sensors

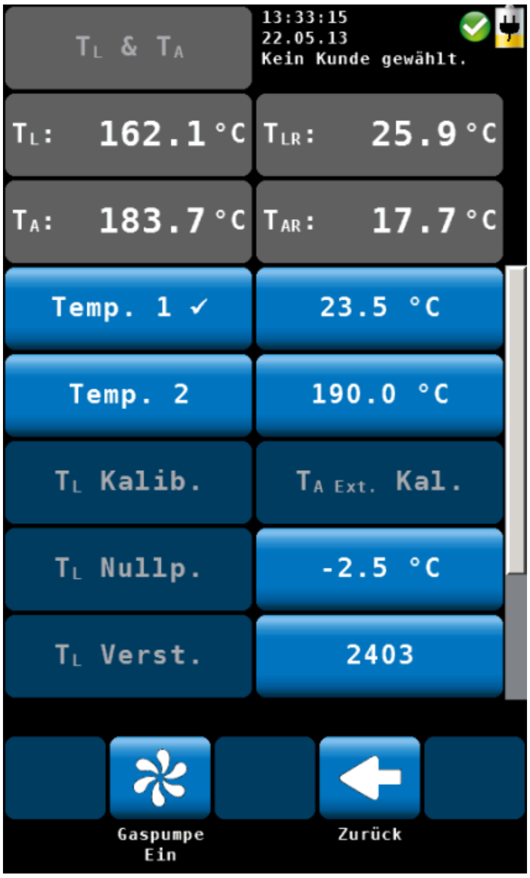


Fig. 4: Calibration of the temperature sensors

- In calibration mode click on **T_L & T_A Kalibrieren**. Plug the T_A probe (Flue gas temperature) and the T_L probe (ambient air temperature) into the point of reference for the lower temperature range (e.g. copper block).
- Click on the value next to **Temp 1** and enter the temperature of the point of reference.
- Click on OK to confirm.
- As soon as the temperature values T_L and T_A are stable (in the upper left side of the screen), click on the **Temp.1** key to save the values. **Temp1** will be marked by a checkmark (see figure 4).
- Plug the T_A probe (Flue gas temperature) and the T_L probe (ambient air temperature) into the point of reference for the upper temperature range (e.g. heating block, T_L max. 100 °C).
- Click on the value next to **Temp 2** and enter this reference temperature value. Click on OK to confirm.
- As soon as the temperature values T_L and T_A are stable (in the upper left side of the screen), click on the **Temp.2** key to save the values. **Temp.2** will be marked by a checkmark.
- Click on **T_L Kalib.** to save the new T_L calibration.
- Click on **$T_{A\ Ext.}$. Kal.** to save the new T_A calibration.
- Click on "Zurück" (Back) to return to the calibration menu.
- Click on **OK** to save the new calibrated values. The question "Änderungen übernehmen?" ("Save modifications?") will appear. Confirm with "Ja" (Yes).

4.5 Calibration of the pressure sensor.

Fig. 5: Negative pressure connector

Fig. 6: upper part of the screen during the pressure calibration

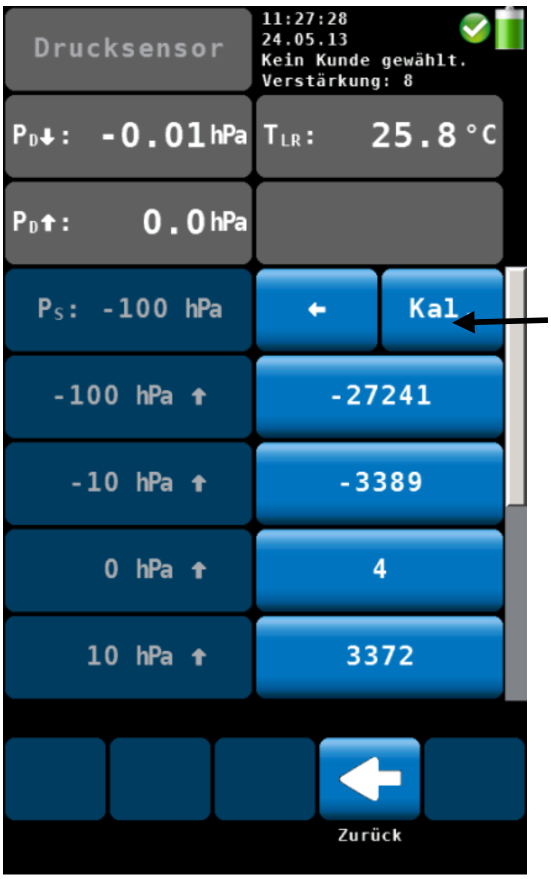


Fig. 7: During the pressure calibration

- In the calibration menu click on **Drucksensor Kalibrieren**.
- Connect the negative pressure connector at the bottom of the Wöhler A 550 with the negative connector of the pressure calibrator.
- Start the pressure calibrator.
- In the screen of the analyzer click on **Kalibrieren Start.**
- Set the pressure calibrator to the reference value **P_S** (-100 hPa).
- As soon as the pressure reading in the upper left part of the analyzer's screen is stable, click on **Kal.**
- - 100 hPa is marked by a checkmark now, and the value -10 hPa can be calibrated.
- Proceed the same way with the other values (-100 hPa, -10 hPa, 0 hPa, +10 hPa, 100hPa).
- Click on "Zurück" (Back) to return to the calibration menu.
- Click on **OK** to save the new calibrated values. The question "Änderungen übernehmen?" ("Save modifications?") will appear. Confirm with "Ja" (Yes).

4.6 Calibration of the O_2 Sensor

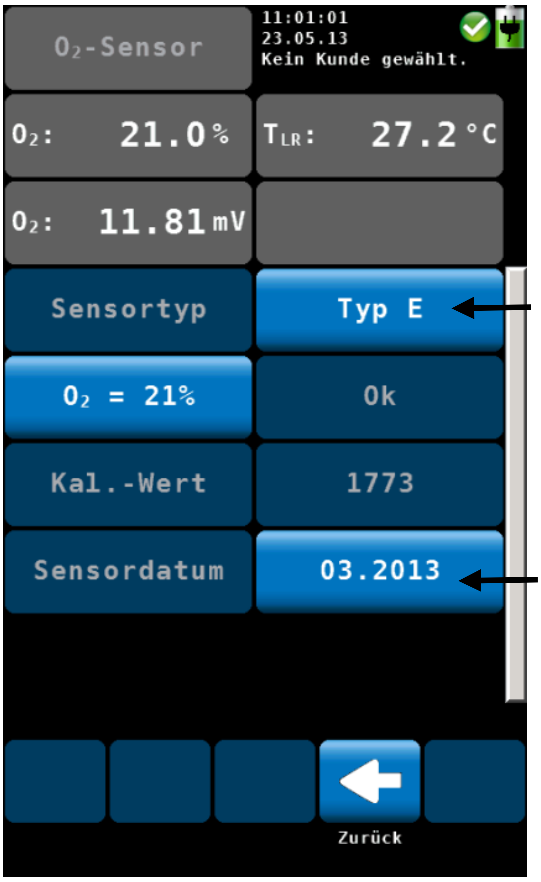


Fig. 8: Selection of the sensor type and entering the date

- In the calibration menu click on **O_2-Sensor Kalibrieren**.

> NOTE!
> If you did not install a new O_2 sensor, you will not have to carry out the O_2 calibration.

- Select the sensor type (normally type E).
- Enter the date when the new sensor has been installed (month and year).
- Click on "Zurück" (Back) to return to the calibration menu.

4.7 Calibration of the CO sensor

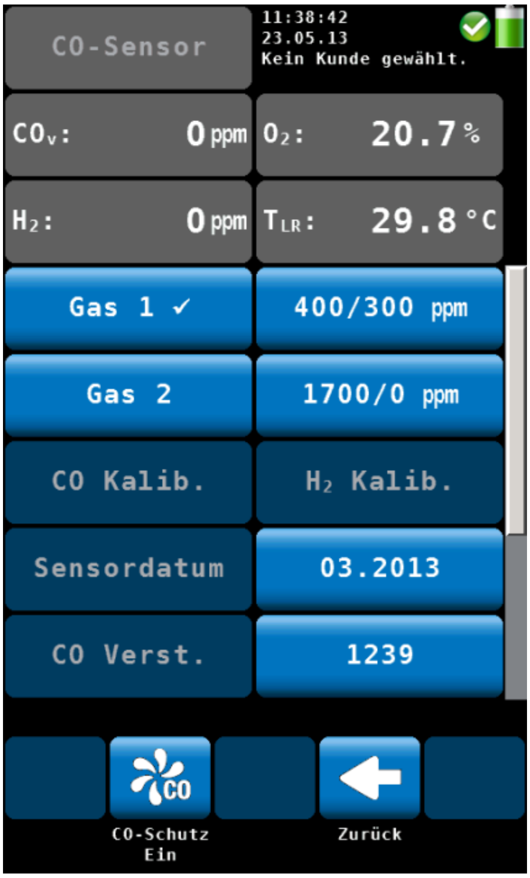


Fig. 9: Calibration of the CO sensor

- In the calibration menu click on **CO-Sensor Kalibrieren**.
- To enter the CO reference value and the H_2 reference value of **Gas 1** proceed as follows: Click on the ppm value next to Gas 1.
- Click on OK to confirm.
- Apply the correspondent test gas to the sensor.
- As soon as the CO_V reading and the H_2 reading respectively are stable (approx. after 3 minutes), click on Gas 1 to save the values. **Gas1** will be marked by a checkmark (see figure on the left).
- Proceed the same way with **Gas 2** (1700/0 ppm).
- Click on **CO Kalib**. to save the new CO calibration.
- Click on **H_2 Kalib**. to save the new H_2 calibration.
- Click on "Zurück" (Back) to return to the calibration menu.
- Click on **OK** to save the new calibrated values. The question "Änderungen übernehmen?" ("Save modifications?") will appear. Confirm with "Ja" (Yes).

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